

DSA Interview Master List (95% Coverage)

Arrays Basics

- Find Maximum and Minimum in an Array
- Reverse an Array
- Rotate Array by K Steps
- Move Zeroes
- Remove Duplicates from Sorted Array
- Best Time to Buy and Sell Stock
- Majority Element
- Merge Sorted Arrays

Sliding Window

- Maximum Sum Subarray of Size K
- Longest Substring Without Repeating Characters
- Longest Subarray with Sum $\leq K$
- Minimum Size Subarray Sum
- Maximum Number of Vowels in a Substring
- Permutation in String
- Longest Repeating Character Replacement
- Subarray Product Less Than K

Two Pointers

- Two Sum II (Sorted Array)
- Valid Palindrome
- Container With Most Water
- Remove Element
- Merge Two Sorted Arrays (In-place)
- Trapping Rain Water
- 3Sum (Conceptual)

Prefix Sum

- Subarray Sum Equals K
- Subarray Sum Divisible by K
- Continuous Subarray Sum
- Zero Sum Subarray (Existence)
- Range Sum Query
- Maximum Length Subarray with Sum K
- Count Subarrays with Equal 0s and 1s

Binary Search

- Binary Search
- First and Last Occurrence
- Search Insert Position
- Find Peak Element

- Search in Rotated Sorted Array
- Find Minimum in Rotated Sorted Array
- Square Root of X
- Koko Eating Bananas

Sorting

- Bubble Sort (Concept)
- Selection Sort (Concept)
- Insertion Sort (Concept)
- Merge Sort
- Quick Sort (Concept)
- Sort Colors

Stack

- Valid Parentheses
- Min Remove to Make Valid Parentheses
- Backspace String Compare
- Next Greater Element
- Daily Temperatures
- Evaluate Reverse Polish Notation
- Largest Rectangle in Histogram

Linked List

- Reverse Linked List
- Middle of the Linked List
- Detect Cycle
- Detect Cycle II
- Remove Nth Node from End
- Merge Two Sorted Lists
- Palindrome Linked List

Trees

- Preorder Traversal
- Inorder Traversal
- Postorder Traversal
- Level Order Traversal
- Maximum Depth of Binary Tree
- Invert Binary Tree
- Diameter of Binary Tree
- Validate BST

Heap

- Kth Largest Element
- Top K Frequent Elements
- K Closest Points
- Merge K Sorted Lists
- Find Median from Data Stream

Greedy

- Activity Selection
- Assign Cookies
- Jump Game
- Gas Station
- Partition Labels
- Minimum Platforms

Graphs

- Number of Islands
- Flood Fill
- BFS Shortest Path
- DFS Traversal
- Connected Components
- Rotten Oranges

Backtracking

- Subsets
- Permutations
- Combination Sum
- Generate Parentheses

Bit Manipulation

- Single Number
- Counting Bits
- Power of Two